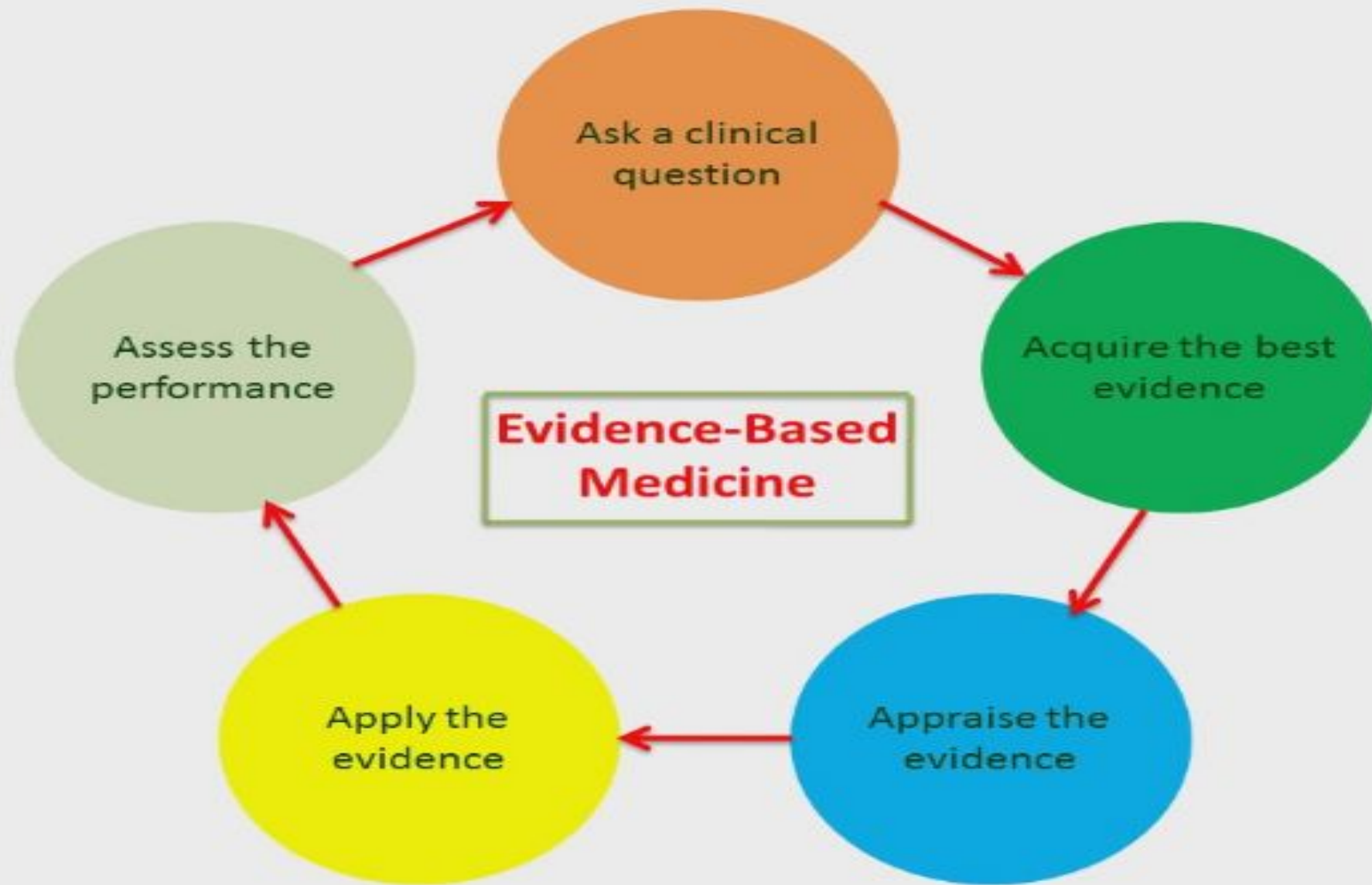


Module 6

Future of Healthcare using AI and ML

The concept of Evidence based medicine

- Evidence-based medicine (EBM) is an approach to medical practice that emphasizes the use of the best available scientific evidence to guide clinical decision-making.
- It is a systematic and rigorous approach that helps healthcare professionals, including doctors, nurses, and other practitioners, make informed decisions about the care of individual patients.



The key components of evidence-based medicine

1. Clinical Expertise:

- EBM acknowledges the importance of clinical expertise and the practitioner's judgment in patient care.
- It doesn't rely solely on scientific evidence but combines it with the practitioner's knowledge and experience.

2. Best Available Evidence:

- EBM places a strong emphasis on using the most current and reliable scientific evidence.
- This evidence typically comes from well-designed clinical studies, such as randomized controlled trials (RCTs), systematic reviews, and meta-analyses.
- These studies are considered more reliable because they follow specific methodologies to reduce bias and ensure the validity of their findings.

The key components of evidence-based medicine

3. Patient Values and Preferences:

- EBM recognizes the importance of considering the individual patient's values, preferences, and unique circumstances when making treatment decisions.
- What works best for one patient may not be suitable for another, and EBM encourages shared decision-making between the healthcare provider and the patient.

4. Integration of Evidence:

- In practice, EBM involves a process of searching for, critically appraising, and integrating the best available evidence into clinical decision-making.
- This often requires staying up-to-date with the latest research and guidelines.

5. Continuous Learning:

- EBM promotes lifelong learning among healthcare professionals.
- Medical knowledge is constantly evolving, so practitioners need to adapt and update their practices based on new evidence.



The process of applying evidence-based medicine steps

1. Formulating a Clinical Question:

- The healthcare provider starts by formulating a clear and focused question about a particular medical issue or treatment.
- This question is typically structured using the PICO(T) framework, which stands for Patient/Population, Intervention, Comparison, Outcome, and Timeframe.

2. Searching for Evidence:

- The practitioner then searches for relevant scientific literature using databases like PubMed, Embase, or Cochrane Library.
- They look for high-quality studies that address the question and provide meaningful insights.

The process of applying evidence-based medicine steps

3.Critical Appraisal:

- The selected studies are critically appraised to assess their methodological quality, potential biases, and relevance to the clinical question.
- This step helps determine the strength of the evidence.

4.Integrating Evidence:

- The practitioner integrates the evidence with their clinical expertise and takes into account the patient's values and preferences to make a treatment decision.

The process of applying evidence-based medicine steps

5.Communication and Shared Decision-Making:

- The healthcare provider discusses the options and their recommendations with the patient, considering the evidence, potential benefits and risks, and the patient's individual circumstances.
- Shared decision-making ensures that the patient is actively involved in their healthcare choices.

The process of applying evidence-based medicine steps

6. **Monitoring and Evaluation:**

- After implementing the chosen treatment plan, healthcare providers should monitor the patient's progress and make adjustments as needed, based on ongoing evidence and the patient's response.

Personalized Medicine in healthcare

- Personalized medicine, also known as precision medicine, is an innovative approach to healthcare that tailors medical decisions, treatments, and interventions to the individual characteristics of each patient.
- This approach recognizes that each person is unique in terms of their genetic makeup, environment, lifestyle, and health history, and aims to use this individualized information to optimize healthcare outcomes.

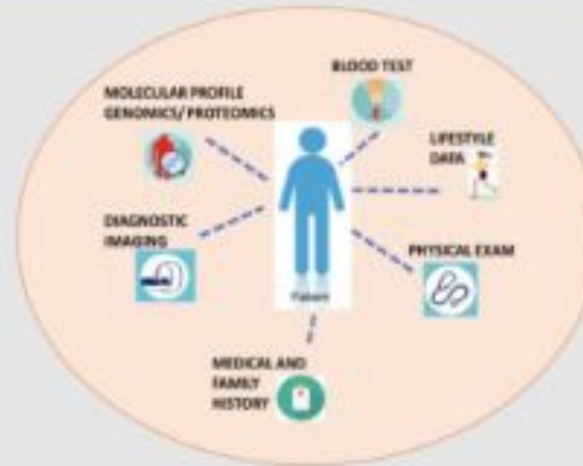
PERSONALISED MEDICINE

PREVENTION



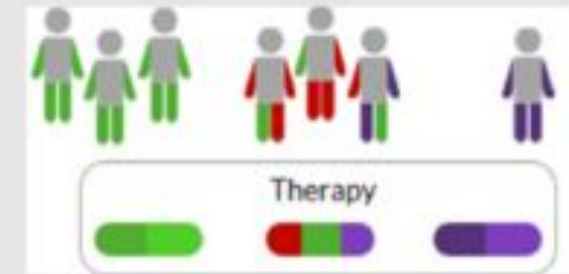
Early detection of patients at risk, Improve preventive measures (individual/collective)

DIAGNOSIS



Accurate disease diagnosis enabling individualized treatment strategy

TREATMENT



Improved outcomes through targeted treatments and reduced side effects

Key elements and concepts for personalized medicine

- 1. Genomic Information:** A central component of personalized medicine is the use of genomic data. This involves analyzing a person's genetic makeup, including variations in their DNA, to understand how these genetic factors influence their health and susceptibility to diseases. Advances in technologies like DNA sequencing have made it increasingly accessible and cost-effective to gather genomic information.
- 2. Targeted Therapies:** Personalized medicine seeks to identify specific genetic or molecular markers associated with diseases. By understanding the genetic basis of diseases, healthcare providers can develop targeted therapies that are more effective and have fewer side effects. For example, certain cancer treatments are now designed to target specific genetic mutations driving the growth of tumors.

Key elements and concepts for personalized medicine

3.Risk Assessment: Personalized medicine can help assess an individual's risk of developing certain diseases based on their genetic profile. This information can guide preventive measures and early interventions to reduce the risk or detect diseases at an earlier, more treatable stage.

4. Drug Response Prediction: Genetic information can help predict how an individual will respond to a particular medication. This can be used to tailor drug selection and dosages to maximize therapeutic benefits while minimizing adverse effects. Pharmacogenomics is a subfield of personalized medicine that focuses on drug-gene interactions.

Key elements and concepts for personalized medicine

5. Disease Prevention and Early Detection: Personalized medicine encourages proactive health management. By identifying genetic predispositions to certain conditions, individuals can make lifestyle changes, undergo regular screenings, and take preventive measures to reduce their disease risk.

6. Treatment Decision Support: In complex medical cases, personalized medicine can provide guidance to healthcare providers about the most appropriate treatment options for a particular patient. This may involve selecting the most effective therapy based on the patient's genetics or avoiding treatments that are unlikely to work.

Key elements and concepts for personalized medicine

- 7. Patient-Centered Care:** Personalized medicine places the patient at the center of healthcare decision-making. It involves shared decision-making between the patient and healthcare provider, with a focus on the patient's values, preferences, and individual circumstances.
- 8. Data Integration:** Personalized medicine requires the integration of diverse data sources, including genomic data, electronic health records, environmental information, and lifestyle data. Advanced analytics and bioinformatics tools are used to interpret and make sense of this complex data.
- 9. Ethical and Privacy Considerations:** The use of genetic and personal data in healthcare raises important ethical and privacy concerns. Protecting patient privacy and ensuring the responsible use of genetic information are critical aspects of implementing personalized medicine.

Connected Medicine

- "Connected Medicine" is not a widely recognized or standardized term in the field of healthcare or medicine.
- However, it's possible that the term has emerged or gained prominence since then, or it could refer to a concept that is better known by a different name.

Connected Medicine

- 1. Connected Health:** This term often refers to the use of technology, such as wearable devices, mobile apps, and remote monitoring tools, to collect and transmit health-related data from patients to healthcare providers. These technologies enable real-time tracking of patients' health status and can improve communication and coordination of care.
- 2. Telemedicine and Telehealth:** Connected Medicine may be a term used to describe the use of telecommunication technologies to deliver medical services remotely. This can include virtual doctor's appointments, remote monitoring of patients with chronic conditions, and the exchange of medical information over secure networks.

Connected Medicine

3. Health Information Exchange (HIE): Connected Medicine could relate to the sharing of patient health information across different healthcare providers and systems. HIE facilitates the seamless exchange of medical records, test results, and other health data, improving the continuity of care.

4. Interoperability: It might refer to efforts to enhance the interoperability of health information systems and electronic health records (EHRs) so that patient data can be easily shared among different healthcare providers, improving care coordination.

Connected Medicine

4. Internet of Things (IoT) in Healthcare: Connected Medicine could involve the use of IoT devices, such as smart medical devices and sensors, to collect and transmit data for remote monitoring and real-time healthcare decision-making.

5. Patient-Centered Care and Engagement: Connected Medicine may also be associated with initiatives aimed at engaging patients more actively in their healthcare through online patient portals, mobile health apps, and other digital tools that connect patients with their healthcare providers.

6. Population Health Management: Connected Medicine could be related to strategies for monitoring and managing the health of entire populations by aggregating and analyzing health data from various sources to identify trends and potential interventions.

Digital Health and Therapeutics

- Digital health and digital therapeutics are related concepts that leverage technology and digital tools to enhance healthcare delivery, improve patient outcomes, and promote overall well-being. While they share some similarities, they have distinct focuses and purposes:

- **Digital Health:**

1. **Definition:** Digital health encompasses a broad range of technologies and solutions that use digital platforms, electronic devices, software, and data to support and enhance healthcare and health-related services. It encompasses various aspects of healthcare, including prevention, diagnosis, treatment, monitoring, and patient engagement.

Digital Health and Therapeutics

1. Examples:

- **Wearable Devices:** Devices like smartwatches and fitness trackers collect health data (e.g., heart rate, activity levels) and provide insights to users.
- **Telemedicine:** Virtual healthcare visits conducted through video calls, phone calls, or secure messaging platforms.
- **Mobile Health (mHealth) Apps:** Mobile applications that offer health-related information, tracking, or assistance (e.g., medication reminders, symptom tracking).
- **Electronic Health Records (EHRs):** Digital systems for storing and managing patient health records.
- **Health Information Exchange (HIE):** Platforms that facilitate the secure exchange of patient health data among healthcare providers.
- **Healthcare Analytics:** Use of data analysis and artificial intelligence to improve healthcare decision-making and population health management.

Digital Health and Therapeutics

1. Goals:

- Improve access to healthcare services, especially in remote or underserved areas.
- Enhance patient engagement and self-management of health.
- Facilitate data sharing and communication among healthcare providers.
- Enable personalized and data-driven healthcare decisions.
- Promote preventive care and early detection of health issues.

Digital Therapeutics

1. **Definition:** Digital therapeutics refer to evidence-based therapeutic interventions that are delivered through software or digital platforms. These interventions are designed to treat, manage, or prevent specific medical conditions or diseases. Unlike general health apps, DTx solutions are rigorously tested for efficacy and safety and often require regulatory approvals.
2. **Examples:**
 - **Diabetes Management Apps:** Digital tools that provide personalized guidance on managing blood sugar levels.
 - **Mental Health Apps:** Software programs designed to treat conditions like depression, anxiety, or insomnia through cognitive-behavioral therapy techniques.
 - **Substance Use Disorder Interventions:** Digital programs to support individuals in overcoming addiction.
 - **Chronic Disease Management:** DTx for conditions like hypertension or chronic obstructive pulmonary disease that help patients adhere to treatment plans and improve outcomes.

Digital Therapeutics

1. Goals:

- Deliver clinically proven treatments or interventions for specific medical conditions.
- Improve patient adherence to treatment plans.
- Offer cost-effective alternatives to traditional therapy or medication.
- Enable remote monitoring and data collection for healthcare providers.
- Tailor interventions to individual patient needs and preferences.

- In summary, digital health is a broad category encompassing all digital technologies and strategies used to support healthcare, while digital therapeutics are a subset of digital health focused on delivering evidence-based treatments or interventions through software or digital platforms. Both play significant roles in modern healthcare by improving access to care, enhancing patient engagement, and leveraging data to improve health outcomes.

Digital Health

All technologies used for health related purposes.
No clinical evidence or regulation needed.

Examples: Telehealth and wellness or lifestyle apps



Digital Medicine

Product used to measure or intervene in healthcare. Must have clinical evidence.

Examples: Wearable sensors that transmit data to healthcare providers



Digital Therapeutics

Products used to prevent, treat or manage disease. Must have clinical evidence and regulatory clearance.

Examples: FDA approved cognitive behavioral therapy apps



DIGITAL HEALTH vs DIGITAL MEDICINE vs DIGITAL THERAPEUTICS

	Digital health	Digital medicine	Digital therapeutics (DTx)
Relation	Includes digital medicine and DTx	Subset of digital health	Subset of digital medicine
Types of products	Different health-related products	Products to measure human health	Products to deliver therapeutic interventions
Clinical evidence	Not always required	Required	Required
Regulatory oversight	Not required	Requirements vary	Always required
Examples	• Electronic health record platforms • Health information portals	• Blood glucose sensors • Heart rate monitoring tools	• Behavior programs • Self-care programs

